General Dessert #1: Safety

RF exposure rules, AC power and wiring safety, grounding, fusing, interlocks, and tower work.

Groups G0A + G0B · 25 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

RF Exposure: Principles

Question	Answer	Q
How RF energy affects body tissue	**It heats body tissue.**	G0A01
What determines RF exposure from a signal	**Duty cycle, frequency, and power density** (all three).	G0A02
How to show compliance with FCC RF exposure rules	**OET Bulletin 65 calculation, computer modeling, or measurement with calibrated equipment** (any of these).	G0A03
What “time averaging” means	**Total RF exposure averaged over a certain period.**	G0A04
Effect of duty cycle on RF exposure	**Lower duty cycle permits greater power levels to be transmitted.**	G0A07
Which stations are subject to FCC RF exposure rules	**All stations with a time-averaged transmission of more than one milliwatt.**	G0A12

RF Exposure: Compliance Actions

Situation	Action	Q
Evaluation shows you exceed permissible RF limits	**Take action to prevent human exposure to the excessive RF fields.**	G0A05
Station fails the FCC RF exposure exemption criteria	**Perform an RF exposure evaluation per FCC OET Bulletin 65.**	G0A06
Required steps to comply with RF safety regs	**Perform a routine RF exposure evaluation and prevent access to identified high-exposure areas.**	G0A08
Instrument to accurately measure RF field strength	**A calibrated field strength meter with a calibrated antenna.**	G0A09
Neighbor may exceed limit from main lobe of beam	**Take precautions so the antenna cannot be pointed in their direction when they are present.**	G0A10
Indoor transmitting antenna installed	**Make sure MPE limits are not exceeded in occupied areas.**	G0A11

AC Power, Wiring, and Fusing

Question	Answer	Q
Which wires get fuses/breakers in a 4-conductor 240 VAC circuit	**Only the hot wires.** (Never fuse the neutral or ground.)	G0B01
Minimum wire size for a 20 A breaker (NEC)	**AWG number 12.**	G0B02
Correct fuse/breaker size for AWG 14 wiring	**15 amperes.**	G0B03
What trips a GFCI	**Current flowing from one or more of the hot wires directly to ground.**	G0B05
What the National Electrical Code covers	**Electrical safety of the station.**	G0B06
Purpose of a power supply interlock	**To ensure dangerous voltages are removed if the cabinet is opened.**	G0B12

Grounding, Lightning, Generators, and Towers

Question	Answer	Q
Lightning protection ground system location	**Outside the building.**	G0B04
Lightning ground-rod bonding requirement	**Bonded together with all other grounds.**	G0B11
Lightning arrestor location	**Where feed lines enter the building.**	G0B13
Emergency generator installation	**Operate in a well-ventilated area.**	G0B09
Lead-tin solder danger	**Lead contamination of food if hands not washed.**	G0B10
Tower climbing with safety harness	**Confirm harness is rated for climber’s weight and within service life.**	G0B07
Before climbing a tower with powered devices	**Lock out and tag all circuits supplying power to the tower.**	G0B08

Dessert 1 Question Bank (G0A + G0B)

G0A01. What is one way that RF energy can affect human body tissue?
A. It heats body tissue
B. It causes radiation poisoning
C. It causes the blood count to reach a dangerously low level
D. It cools body tissue

G0A02. Which of the following is used to determine RF exposure from a transmitted signal?
A. Its duty cycle
B. Its frequency
C. Its power density
D. All these choices are correct

G0A03. How can you determine that your station complies with FCC RF exposure regulations?
A. By calculation based on FCC OET Bulletin 65
B. By calculation based on computer modeling
C. By measurement of field strength using calibrated equipment
D. All these choices are correct

G0A04. What does “time averaging” mean when evaluating RF radiation exposure?
A. The average amount of power developed by the transmitter over a specific 24-hour period
B. The average time it takes RF radiation to have any long-term effect on the body
C. The total time of the exposure
D. The total RF exposure averaged over a certain period

G0A05. What must you do if an evaluation of your station shows that the RF energy radiated by your station exceeds permissible limits for possible human absorption?
A. Take action to prevent human exposure to the excessive RF fields
B. File an Environmental Impact Statement (EIS-97) with the FCC
C. Secure written permission from your neighbors to operate above the controlled MPE limits
D. All these choices are correct

G0A06. What must you do if your station fails to meet the FCC RF exposure exemption criteria?
A. Perform an RF exposure evaluation in accordance with FCC OET Bulletin 65
B. Contact the FCC for permission to transmit
C. Perform an RF exposure evaluation in accordance with World Meteorological Organization guidelines
D. Use an FCC-approved band-pass filter

G0A07. What is the effect of modulation duty cycle on RF exposure?
A. A lower duty cycle permits greater power levels to be transmitted
B. A higher duty cycle permits greater power levels to be transmitted
C. Low duty cycle transmitters are exempt from RF exposure evaluation requirements
D. High duty cycle transmitters are exempt from RF exposure requirements

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G0A08. Which of the following steps must an amateur operator take to ensure compliance with RF safety regulations?
A. Post a copy of FCC Part 97.13 in the station
B. Notify neighbors within a 100-foot radius of the antenna of the existence of the station and power levels
C. Perform a routine RF exposure evaluation and prevent access to any identified high exposure areas
D. All these choices are correct

G0A09. What type of instrument can be used to accurately measure an RF field strength?
A. A receiver with digital signal processing (DSP) noise reduction
B. A calibrated field strength meter with a calibrated antenna
C. An SWR meter with a peak-reading function
D. An oscilloscope with a high-stability crystal marker generator

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G0A10. What should be done if evaluation shows that a neighbor might experience more than the allowable limit of RF exposure from the main lobe of a directional antenna?
A. Change to a non-polarized antenna with higher gain
B. Use an antenna with a higher front-to-back ratio
C. Take precautions to ensure that the antenna cannot be pointed in their direction when they are present
D. All these choices are correct

G0A11. What precaution should be taken if you install an indoor transmitting antenna?
A. Locate the antenna close to your operating position to minimize feed-line radiation
B. Position the antenna along the edge of a wall to reduce parasitic radiation
C. Make sure that MPE limits are not exceeded in occupied areas
D. Make sure the antenna is properly shielded

G0A12. What stations are subject to the FCC rules on RF exposure?
A. All commercial stations; amateur radio stations are exempt
B. Only stations with antennas lower than one wavelength above the ground
C. Only stations transmitting more than 500 watts PEP
D. All stations with a time-averaged transmission of more than one milliwatt

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G0B01. Which wire or wires in a four-conductor 240 VAC circuit should be attached to fuses or circuit breakers?
A. Only the hot wires
B. Only the neutral wire
C. Only the ground wire
D. All wires

G0B02. According to the National Electrical Code, what is the minimum wire size that may be used safely for wiring with a 20-ampere circuit breaker?
A. AWG number 20
B. AWG number 16
C. AWG number 12
D. AWG number 8

G0B03. Which size of fuse or circuit breaker would be appropriate to use with a circuit that uses AWG number 14 wiring?
A. 30 amperes
B. 25 amperes
C. 20 amperes
D. 15 amperes

G0B04. Where should the station’s lightning protection ground system be located?
A. As close to the station equipment as possible
B. Outside the building
C. Next to the closest power pole
D. Parallel to the water supply line

G0B05. Which of the following conditions will cause a ground fault circuit interrupter (GFCI) to disconnect AC power?
A. Current flowing from one or more of the hot wires to the neutral wire
B. Current flowing from one or more of the hot wires directly to ground
C. Overvoltage on the hot wires
D. All these choices are correct

G0B06. Which of the following is covered by the National Electrical Code?
A. Acceptable bandwidth limits
B. Acceptable modulation limits
C. Electrical safety of the station
D. RF exposure limits of the human body

G0B07. Which of these choices should be observed when climbing a tower using a safety harness?
A. Always hold on to the tower with one hand
B. Confirm that the harness is rated for the weight of the climber and that it is within its allowable service life
C. Ensure that all heavy tools are securely fastened to the harness
D. All these choices are correct

G0B08. What should be done before climbing a tower that supports electrically powered devices?
A. Notify the electric company that a person will be working on the tower
B. Make sure all circuits that supply power to the tower are locked out and tagged
C. Unground the base of the tower
D. All these choices are correct

G0B09. Which of the following is true of an emergency generator installation?
A. The generator should be operated in a well-ventilated area
B. The generator must be insulated from ground
C. Fuel should be stored near the generator for rapid refueling in case of an emergency
D. All these choices are correct

G0B10. Which of the following is a danger from lead-tin solder?
A. Lead can contaminate food if hands are not washed carefully after handling the solder
B. High voltages can cause lead-tin solder to disintegrate suddenly
C. Tin in the solder can “cold flow,” causing shorts in the circuit
D. RF energy can convert the lead into a poisonous gas

G0B11. Which of the following is required for lightning protection ground rods?
A. They must be bonded to all buried water and gas lines
B. Bends in ground wires must be made as close as possible to a right angle
C. Lightning grounds must be connected to all ungrounded wiring
D. They must be bonded together with all other grounds

G0B12. What is the purpose of a power supply interlock?
A. To prevent unauthorized changes to the circuit that would void the manufacturer’s warranty
B. To shut down the unit if it becomes too hot
C. To ensure that dangerous voltages are removed if the cabinet is opened
D. To shut off the power supply if too much voltage is produced

G0B13. Where should lightning arrestors be located?
A. Where the feed lines enter the building
B. On the antenna, opposite the feed point
C. In series with each ground lead
D. At the closest power pole ground electrode

Answer Key

G0A01: A	G0A02: D	G0A03: D	G0A04: D	G0A05: A
G0A06: A	G0A07: A	G0A08: C	G0A09: B	G0A10: C
G0A11: C	G0A12: D	G0B01: A	G0B02: C	G0B03: D
G0B04: B	G0B05: B	G0B06: C	G0B07: B	G0B08: B
G0B09: A	G0B10: A	G0B11: D	G0B12: C	G0B13: A